Tangents and normals

1 For each of the following curves:

i Find the derivative of the function.

ii Find the gradient of the tangent at the given point.

a $y = \ln 2x$ at the point where $x = 5$.

b $y = 4x + \ln 3x$ at the point where $x = \frac{1}{5}$.

c $y = \ln\left(x^2 + 5\right)$ at the point where $x = 3$.

d $y = \ln\left(x^4 + 4\right)$ at the point where $x = 2$.

e $y = x^2 \ln x^2$ at the point where $x = e$.

2 For each of the following curves:

i Find the exact gradient of the normal to the curve.

ii Find the equation of the normal to the curve.

a $y = \ln(x^4 + x)$ at the point $(1, \ln 2)$.

b $y = \ln x$ at the point where $x = 5$.

c $y = \log_3 x$ at the point where $x = 3$.

3 Consider the function $y = \ln x$.

a Find the x-intercept.

b Find the equation of the tangent to the curve at the point where it crosses the x-axis.

c Find the equation of the normal to the curve at the point where it crosses the x-axis.

4 Consider the function $y = \ln(-x)$.

a State the domain of this function.

b Find $\frac{dy}{dx}$.

c Hence, find the gradient of the tangent to the curve at $x = -5$.

5 Consider the function $y = 2\ln\left(x^2 + e\right)$.

a Find the derivative of y.

b Evaluate the derivative at $x = 0$.

c Hence, find the equation of the tangent to the curve at $x = 0$.

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6 Consider the function $y = \ln(\ln x)$.

a Find the derivative of y.

b Find y when $x = e$.

c Hence, find the equation of the tangent to the curve at $x = e$.

7 Consider the function $y = 3\ln\left(x^3 + 2\right)$.

a Find the derivative of y.

b Evaluate the derivative at $x = 1$.

c Hence, find the equation of the normal to the curve at $x = 1$.

8 Consider the function $y = -2(\ln 3x^2 - x)$.

a Find the derivative of y.

b Find y when $x = 2$, correct to two decimal places.

c Find the equation of the normal to the curve at $x = 2$.

9 At the point (a, b) on the curve $y = \ln(-2x)$, the gradient of the tangent to the curve is $-\frac{1}{3}$. Solve for the value of a.

Graphs of logarithmic functions

10 For each of the following functions:

i State the domain of y.

ii Find y'.

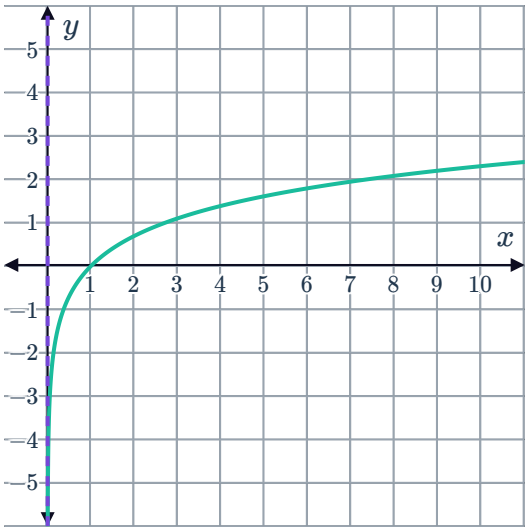
iii State the domain of y'.

a $y = \ln(-3x)$

b $y = \ln(x - 7) + \ln x$

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11 Consider the graph of $y = \ln x$.



a State whether the function is increasing or decreasing.

b State whether the gradient to the curve is negative at any point on the curve.

c Describe the change in the gradient of the tangent as x increases.

d Describe the change in the gradient of the tangent as x gets closer and closer to 0.

12 Consider the function $y = \ln\left(\dfrac{1}{(x-4)^3}\right)$.

a Simplify the function.

b Find $\dfrac{dy}{dx}$.

c State the values of x for which the function is undefined.

d Determine whether the function is strictly increasing, strictly decreasing, or neither. Explain your answer.

13 Consider the function $f(x) = \ln x^4$.

a Find $f'(x)$. **b** Find $f'(2)$. **c** Find $f''(x)$. **d** Find $f''(2)$.

e State whether the function is increasing or decreasing at $x = 2$.

f Describe the concavity of the function at $x = 2$.

14 Consider the function $f(x) = x^2 \ln x$.

a State the domain of this function.

b Find the exact coordinates of the turning point.

c Determine the nature of the turning point.

15 Consider the function $f(x) = \dfrac{\ln x}{x}$ for $x > 0$.

a Find the x-coordinate of the stationary point.

b State whether the stationary point is a maximum or minimum.

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16 Consider the function $y = x \ln x$.

a State the domain of the function.

b Find y'.

c Determine the exact coordinates of the turning point.

d Find y'' at the turning point.

e State whether the turning point is a maximum or minimum value of the function.

f State the range of the function in exact form.

g Describe the behaviour of the function as $x \to \infty$.

h Sketch the graph of the function.

17 Consider the function $y = x - \ln x$.

a Find y'.

b State the coordinates of any stationary points.

c Find y''.

d Determine the nature of the stationary points.

e Describe what happens to the function as $x \to \infty$.

f Describe what happens to the function as $x \to 0$.

g Sketch the graph of the function.

18 Consider the function $f(x) = (\ln x - 1)^3$.

a Find any x-intercepts.

b Find the exact coordinates of any stationary points.

c Find $f''(x)$.

d Complete the following table:

x	1	e	e^3	e^4
$f''(x)$				

e Hence state the coordinates of any points of inflection.

f Sketch the graph of the function.

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19 Consider the functions $f(x) = \ln x$ and $g(x) = \ln 3x$.

- **a** Sketch both functions on the same Cartesian plane.
- **b** Find $f'(x)$.
- **c** Find $g'(x)$.
- **d** What can be concluded about the tangents of the curves at any given x-value?

Applications

20 A plane takes off from an airport at sea level and its altitude h, in metres, t minutes after taking off, is given by $h = 600 \ln(t + 1)$.

- **a** Write an expression for the rate at which the plane ascends, exactly t minutes after taking off.
- **b** Hence, find the rate of ascent at exactly 4 minutes after take off.
- **c** Determine whether the plane is ascending at an increasing or decreasing rate.

21 A new airline wants to forecast its predicted growth in passenger numbers from now until the end of the decade.
In the current month $(t = 0)$, the airline services 3400 passengers. The forecasters use the following function to model the number of passengers t months from the current month, for $t \geq 0$ and some constants m and k:

$$P(t) = k \ln(t + 1) + m$$

- **a** State the value and explain the meaning of the constant m.
- **b** Find k if 4 months later, the airline services 3641 passengers. Round your answer to the nearest whole number.
- **c** Find t, the whole number of months it will take for the airline to reach 3745 passengers.
- **d** Hence, find the initial rate $P'(0)$, at which the number of passengers is increasing.
- **e** Find the number of months, t, for the rate of increase to be half the initial rate of increase.

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22 Before conservationists release two different species of endangered birds back into the wild, they introduce some of them into a large enclosure and track their numbers over 20 months.
The populations of species A and B, t months after being in the enclosure, are given by the following functions:

- Species A: $A(t) = 17\ln(8t+5)$
- Species B: $B(t) = 17\ln(6t+25)$

a State the domain of both functions.

b After 1 year, at what rate is the population of species A increasing?

c Calculate the number of whole months, t, in the enclosure, for the populations of species A and B to reach the same level.

d At the time when the populations of the two species are the same, find the exact rate of increase in the population of:

i Species A **ii** Species B

e Hence, state which species will have a greater population after 20 months.

23 Luigi's farm currently produces 10.1 tonnes of barley annually. Over an extended period of drought, he has found that the productivity of his land is decreasing but the rate of decrease is slowing down.
He decides that he will keep his barley farm until annual productivity reaches 0, so he uses the following logarithmic function to model the annual productivity of his land, t years from now:

$$P(t) = A + k\ln(t+1)$$

Note that $P(t)$ is the annual productivity in tonnes after t years, and $P(0)$ is the annual rate of productivity at the start of the model.

a Find A.

b Hence, find the value of k in the model if the annual productivity drops to 7 tonnes after one year. Round your answer to one decimal place.

c Find the rate at which the annual productivity changes 4 years from now, correct to one decimal place.

d Find t, the number of years from now that Luigi will sell his farm. Assume that he only sells at the end of the year.

e Determine the rate at which the annual productivity is decreasing at the end of the last year that Luigi runs the farm. Round your answer to two decimal places.

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